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ORIGINAL RESEARCH

From Practice To Performance: Predicting Soccer Match Outcomes from Training Data

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Abstract

This study analyzes performance data from training sessions of male soccer players to predict their match performance. Regression analysis was conducted to examine the relationship between training metrics—such as energy expenditure, sprint count, and power plays—and match performance features. Additionally, a binary classification model was employed to determine whether a player's match performance could be predicted based on their training sessions leading up to the game. A correlation analysis using multivariate regression further explored the relationship between training session data and match outcomes.

To support training optimization, a drill selection application was developed to maintain a database of past drills, including metrics such as average player load, intensity, and duration. For each of the two training sessions prior to a match, the most relevant drill was identified based on the coach's criteria, ensuring that training exercises closely aligned with match demands.

The findings of this study provide insights into how varying training loads impact match performance and offer a data-driven approach for coaches to refine training sessions, optimize player development, and make informed decisions regarding starting lineups and substitutions.

Keywords Performance Prediction · Soccer Analytics · Match Performance

Introduction

The increasing interest in sports analytics over the last two decades can be attributed to advances in technology where data has been used by teams and individuals to gain a competitive advantage. Statistics have always played a role in sport, but the use of predictive analysis has increased in recent years. The volume of data collected for each game makes a big data problem since it is not readily feasible to gain meaningful insights from raw data. Data driven decision making is being incorporated in different aspects of sports from gambling, fantasy leagues, to improving team dynamics and performance, decision making, pre-venting injuries, etc.

Soccer is a high intensity sport involving various complex factors that allow a team to come together and compete. At the highest level, consistent optimized player performance is key to a successful team. However, there are many factors that affect the physical readiness of players to withstand the demands of competing. The main factor we will be focusing on is the intensity and volume of training sessions leading up to matches. Ultimately, the responsibility of monitoring this workload falls to the coaching staff. Coaching style, organization, discretion, and intentionality of the staff all can affect a team's ability to perform to maxi-mum capacity.

In this research, we focus on how coaching staff can optimize player and team performance using predictive analysis and machine learning. We studied predicting how the player would perform during a match based on their training performance leading up to the match. To support our coaching team, we developed a soccer training application that provides drill suggestions based on the coaches' criteria on intensity and duration (see Fig. 1).

We analyze one drill from each of the two training sessions prior to a match, selecting the drill most similar to actual match conditions, along with performance data for each player. Using this data, we developed a predictive

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Fig. 1 Soccer training application –training sessions drills based on coach's input

model to estimate individual match performance based on their training. This research will help coaches understand how the demands of training sessions impact match day performance and assist in determining the optimal starting lineup and substitution strategies to maximize team performance.

Literature Review

The application of machine learning in sports analytics and prediction analysis is becoming increasingly popular. Specifically, in professional soccer, studies are employing various modeling techniques to predict a myriad of success-related parameters.

Predictive modeling in soccer aims to estimate the probabilities of possible outcomes (win, draw, lose) for a given match. The low scoring nature of the soccer game makes accurate score prediction challenging. Soccer analytics research falls in many categories such as predictive modeling of match outcomes [1–6], match performance based on training workload [7–9] and player performance specific to player role [10, 11].

Elmiligi and Saad (2022) analyzed the returns of 205,182 soccer matches of multiple teams over the period of 2000–2017. In their study they combined machine learning with statistical analysis to enhance the predictive accuracy of soccer match outcomes [1].

The study by Guerrero-Calderón et al. (2021) analyzed the relationship between the training load applied throughout the week and players' performance metrics in games [2]. As a result, match performance can be predicted based on training load during the previous training week. Training total distance negatively affected match physical output while training high intensity positively affected match output.

Joseph, Fenton, and Neil (2006) utilized Bayesian networks for predicting match outcomes [3]. They studied the importance of updating probabilities as new data becomes available. Hvattum and Arntzen (2010) explored the use of feature-based models, using logistic regression to predict match results [4]. Their work highlighted the effectiveness of simple statistical features in making predictions. Recent studies have focused on ensemble methods and more sophisticated models to enhance prediction accuracy. Berrar et al. (2019) investigated the use of ensemble learning, combining multiple machine learning models to improve predictive performance in soccer matches [5]. Kahn (2022) explored deep learning approaches, specifically recurrent neural networks (RNNs) and long short-term memory (LSTM) networks, to model the sequential nature of match events and predict outcomes with higher accuracy [6]. A study by Santos et al. (2021) emphasized the importance of integrating biometric data, including heart rate variability (HRV) and sleep quality, into training load management [7]. They showed that personalized training programs based on these data points could significantly enhance match performance by optimizing recovery and readiness. Research by Miller et al. (2022) highlighted the role of wearable technology in providing real-time data on player workload and physiological responses [8]. This study demonstrated how real-time analytics could be used to adjust training intensity, reducing injury risk and improving performance outcomes. A study by the study by Zhang and Lee (2022) focused on leveraging machine learning algorithms to predict player responses to various training loads [9]. They developed predictive models that could forecast individual players' performance and risk of injury.

Furthermore, research is being conducted on soccer player performance specific to the role they play on their team. Lago-Peñas and Gómez-Ruano (2016) conducted a systematic review focused on the assessment of offensive performance in soccer [10]. Their study used various statistical methods and performance indicators to evaluate offensive play in soccer matches. The research done by Carpita et al. (2021), used statistical analysis to determine the most impactful and effective players in each role [11]. Their methodology assesses team performance based on players' positions and contributions.

Data

The data used is on division I 2022–2024 seasons of one college men's soccer team. Our data consists of 45 matches and 90 trainings session. This data is directly related to physical performance and gathered using PlayerTek technology and platform. PlayerTek is a professional grade sensor device and web-based software that is used to measure and track athletes' physical performance [12]. It uses a GPS system and a range of smart sensors to gather a series of metrics. These sensors and systems are specifically designed for high-level soccer player use.

PlayerTek categorizes the performance metrics into two groups – volume and intensity. Within the volume category, we are analyzing distance, sprint distance, energy, power plays, and sprints. Within the intensity category, we are analyzing top speed, distance per minute, power score, work ratio, and player load. Additional attributes are considered including total duration, duration within speed zone 2–5 and session player score. Session player score is an indication of how much load a player experienced compared to that typical of a full match [13]. PlayerLoad uses triaxial accelerometry data to calculate the load or activity level of the

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player and is the number one metric used by performance coaches in professional and collegiate sport. Research has shown that PlayerLoad is a valid and reliable measure and can provide important insights into player activity. As an example, if a player received a session player score of 75, this means that they experienced a load equal to 75% of a full match load. Further detail and description of each of these attributes can be seen in Table 1.

Methodology

In this study, we introduce a novel approach to soccer analytics by predicting player match performance based on their training sessions leading up to the game. Our methodology involves analyzing training and match performance data to determine key predictors of match success. We employ a combination of regression analysis and classification techniques to identify patterns, along with a practical application to assist coaches in drill selection. While our primary focus is on physical performance metrics, we acknowledge that psychological and external factors, such as confidence, emotional state, opponent skill level, and home vs. away conditions, may also influence player performance but are challenging to quantify. Additionally, some GPS data comes from players who did not participate in the match, introducing variability in the dataset. Our analysis consists of three key stages:

- **Regression Analysis**: We use regression techniques to examine the relationship between training metrics and match performance features. This allows us to identify which training variables, such as energy expenditure, sprint count, and power plays, have the most significant impact on match performance.
- **Binary Classification**: A classification model is applied to determine whether match outcomes can be predicted based on training data. This provides insights into the extent to which training performance influences game results.
- **Correlation Analysis**: A multivariate regression analysis was performed to determine the correlation between the two training sessions and the match data.
- **Drill Selection Application**: We developed an application that maintains an inventory of past training drills along with their average player load. By comparing training data with actual match performance, this tool enables coaches to select optimal drills based on intensity and duration parameters.

Typically, there are 3–4 training sessions per week; however, due to varying match schedules, we standardized our training data by focusing on the two most recent sessions before each match to predict player performance. Each session consists of various drills, beginning with a warm-up and ending with a cool-down. On average, 4–5 different drills are incorporated per session, including passing, dribbling, pressing, and other match related exercises. For each training session, we select the drill that most closely resembles actual match conditions.

To further refine our analysis, we leveraged historical player load data from training sessions to categorize drills based on their similarity to actual match demands. This categorization allows us to identify the training activity that most closely mirrors in-game conditions. Based on specific input from coaches, our soccer training application provides drill recommendations. During each practice session, we extract data from the drill that closely aligns with match conditions for each player, using this information to predict their game performance.

A key challenge in training analysis is the difficulty of isolating different drills within a session. As shown in Fig. 2, training sessions often blend multiple drills, making it hard to attribute specific performance metrics to individual activities. However, by incorporating our soccer training

Table 1 Feature description

Attribute	Description
Training Player score	Percentage of experienced load compared to that of a typical match
Duration	Total time in seconds of the training
Duration within zone 2–5	Total time in seconds of the training the player is moving with velocity greater than 1 m per second
Distance	Total distance covered over the duration of a training
Sprint distance	Distance covered while running at speeds greater than 5 m per second
Power plays	Significant action where power output is over 20 watts per kilogram of body weight
Energy	Amount of energy expended in the duration of a training
Sprints	Total number of sprints indicated by change in speed from below to greater than 5 m per second
Top speed	Maximum speed sustained for at least half a second
Distance per minute	Meters per minute, indicates how hard and intensely a player worked
Power score	Measures power output used per kilogram of player weight
Work ratio	Percentage of time player is working compared to resting, defined as speeds greater than 1.5 m per second
Player load	Measure of instantaneous rate of change of acceleration divided by a scaling factor

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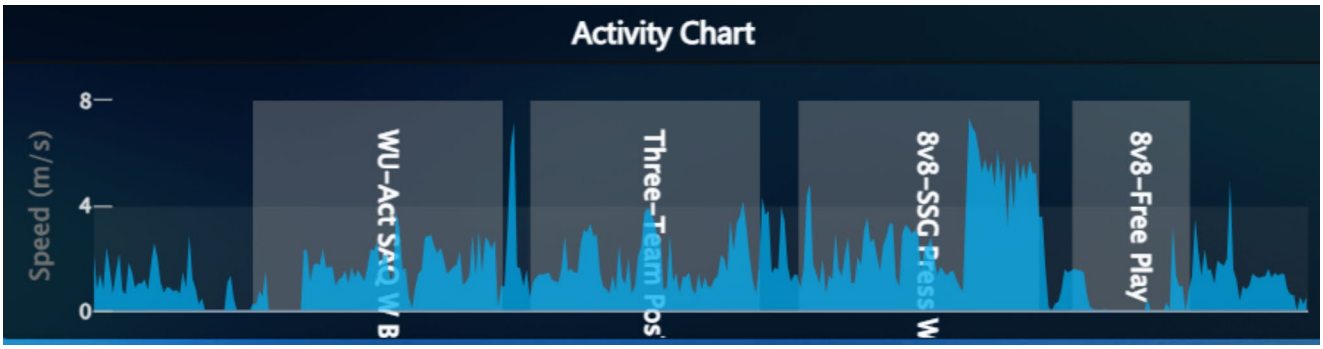

Fig. 2 Activity chart of the training session split for each drill

Table 2 Training features that have the highest correlation with match feature

Match Feature	Training Session Feature	*p*-value
Distance	Power Score	0.02152
	Sprint Distance	0.05166
	Number of Sprints	0.05193
Sprint Distance	Sprint Distance	0.01978
	Power Score	0.02097
	Sprints	0.02143
Number of Sprints	Sprint Distance	0.00552
	Sprints	0.00616
	Power Score	0.00730
Player Load	Power Score	0.02952
	Sprints	0.04340
	Sprint Distance	0.05251
Power Play	Sprint Distance	0.01869
	Power Score	0.01982
	Sprints	0.02093

application, we resolved this issue by precisely tracking the start and end times of each drill, enabling accurate data extraction and analysis.

Regression Analysis

For this study, we analyzed the two training sessions leading up to a match. The first session, held two days before the game, is typically more intense, while the second session, conducted 24 h before kickoff, focuses on lower-intensity activities for recovery. We then performed a regression analysis to assess the impact of various training features, including total distance covered, sprint distance, top speed, and power score, on match performance. The most significant predictors, along with their corresponding p-values, are summarized in Table 2.

Binary Classification

After dividing the training session into specific time segments, we extract only the data from the selected drill for that day to ensure a stronger correlation with the match performance. This approach filters out noise from warm-up, cool-down, and player development drills.

We built machine learning models to predict the match performance based on the features from the training sessions. In our binomial classification model, the 'true class' refers to players outperforming their performance in training sessions. Meaning that their actual player load would be higher than the one that was predicted using the training data. By utilizing these predictions, coaches can learn how to optimize player performance to achieve the best outcomes. We utilized supervised learning to train the model and evaluated the accuracy for each pattern recognition model and the top 4 models are described in Table 3.

Generalized Linear Model (GLM) gave the highest accuracy of 85.7%.

Correlation Analysis

A multivariate regression analysis was conducted to examine the correlation between training sessions and match performance data. The results indicate a positive relationship between training duration, distance run in training, and

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Table 3 Different model methods with accuracy and total time

Model	Accuracy	Std	Total Time	Training Time	Scoring Time
Generalized Linear Model	85.7%	±3.2%	335 ms	278 ms	195 ms
Naive Bayes	75.3%	±6.3%	563 ms	166 ms	163 ms
Fast Large Margin	77.9%	±9.3%	981 ms	178 ms	150 ms
Deep Learning	81.2%	±7.7%	738 ms	1 s	200 ms

distance covered during a match. Specifically, for every additional unit (e.g., kilometer or mile) of distance covered in training, the expected match distance increases by 0.2487 units, assuming other factors in the model remain constant.

Furthermore, training duration and distance run positively correlate with all key match performance metrics, including total distance covered, sprint distance, power plays, number of sprints, player load, and top speed. This suggests that higher training volume generally leads to improved match performance.

However, the relationship between training energy expenditure and match top speed presents an inverse correlation. The regression model shows a negative coefficient of -0.0419, indicating that increased energy use during training is associated with lower top speed during matches. This negative correlation suggests that high energy expenditure in training may contribute to fatigue, limiting a player's top speed during competition.

To enhance match performance, adjustments in training intensity or recovery strategies may be necessary. Players could benefit from a more balanced or strategic approach to energy use in training to prevent excessive fatigue, particularly in areas critical to performance, such as sprinting and maintaining top speed during matches.

Conclusion

This study introduced a soccer training application designed as a comprehensive database of training drills, storing key metrics such as average player load, intensity, and other performance indicators. The application enables coaches to access and filter drills based on specific training needs. By inputting session requirements, such as intensity, duration, or focus areas, coaches receive a curated set of drills that best align with their objectives. Additionally, the Drill Database allows coaches to retrieve detailed information on any drill at any time, without needing to generate a session first.

The research developed a predictive model to assess individual player performance in a soccer match. By analyzing performance metrics from the two training sessions leading up to a match, we identified which players were likely to outperform their training performance during the game. The findings provide a foundation for coaches to optimize their starting lineup and player substitutions throughout the match based on data driven performance predictions.

Regression analyses revealed that power score, sprint distance, and sprint count are key predictors of match performance. Players with higher power output and greater sprinting activity tend to cover more distance and engage in more power plays during matches. Additionally, training duration and distance run showed strong positive correlations with match distance and sprint performance. However, an inverse relationship was found between excessive energy expenditure in training and match top speed, suggesting a potential trade-off between training intensity and peak match performance. These results emphasize the need to optimize training intensity, sprinting capacity, and power output while strategically managing energy expenditure to enhance overall match performance.

While GPS-based performance metrics provide valuable insights, predicting a player's match performance requires more than just numerical summaries of GPS values. Some aspects of match performance, such as game intensity and situational involvement, cannot be fully replicated in training. If training perfectly mimicked match conditions, players would not be adequately prepared for the unique demands of actual competition.

Future Work

Future research will explore the relationship between wellness, training, and match performance [14]. A custom wellness tracking app and form have been developed for the men's soccer team to collect data on sleep quality, fatigue, soreness, stress, energy levels, and muscle pain. The goal is to use these insights to minimize injury risk and enhance performance. Initial findings suggest a strong correlation between training intensity and wellness variables, but further research with a larger dataset is needed to validate these findings and refine injury prevention strategies.

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Data Availability Data will be provided upon request.

Declarations

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